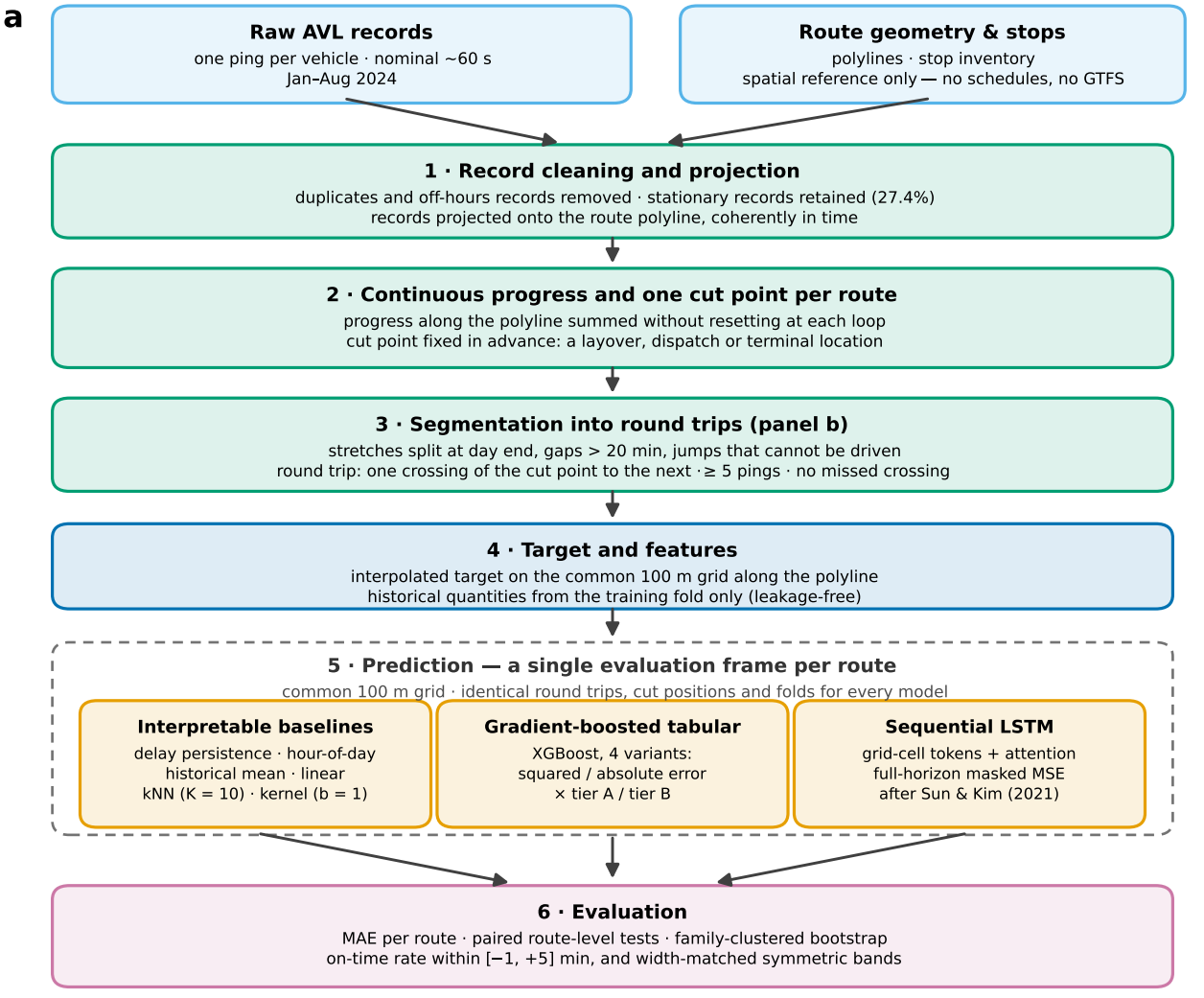
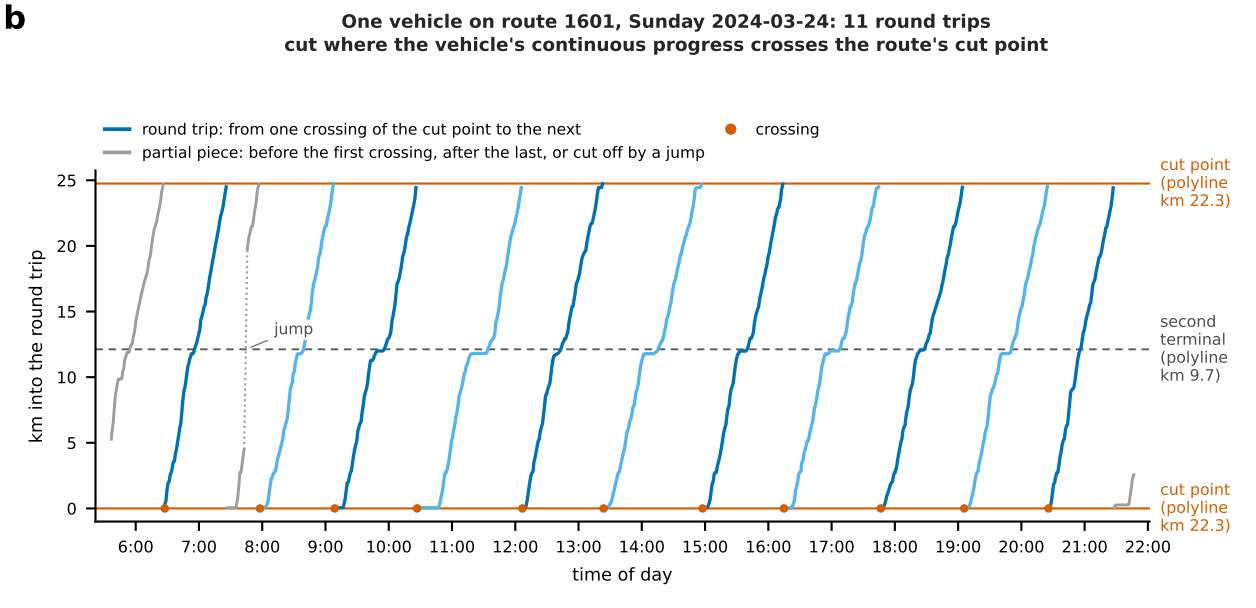

a
Raw AVL records
one ping per vehicle · nominal ~60 s
Jan–Aug 2024
Route geometry & stops
polylines · stop inventory
spatial reference only — no schedules, no GTFS
1 · Record cleaning and projection
duplicates and off-hours records removed · stationary records retained (27.4%)
records projected onto the route polyline, coherently in time
2 · Continuous progress and one cut point per route
progress along the polyline summed without resetting at each loop
cut point fixed in advance: a layover, dispatch or terminal location
3 · Segmentation into round trips (panel b)
stretches split at day end, gaps > 20 min, jumps that cannot be driven
round trip: one crossing of the cut point to the next · ≥ 5 pings · no missed crossing
4 · Target and features
interpolated target on the common 100 m grid along the polyline
historical quantities from the training fold only (leakage-free)
5 · Prediction — a single evaluation frame per route
common 100 m grid · identical round trips, cut positions and folds for every model
Interpretable baselines
delay persistence · hour-of-day
historical mean · linear
kNN (K = 10) · kernel (b = 1)
Gradient-boosted tabular
XGBoost, 4 variants:
squared / absolute error
× tier A / tier B
Sequential LSTM
grid-cell tokens + attention
full-horizon masked MSE
after Sun & Kim (2021)
6 · Evaluation
MAE per route · paired route-level tests · family-clustered bootstrap
on-time rate within [−1, +5] min, and width-matched symmetric bands


b
One vehicle on route 1601, Sunday 2024-03-24: 11 round trips
cut where the vehicle's continuous progress crosses the route's cut point
round trip: from one crossing of the cut point to the next
partial piece: before the first crossing, after the last, or cut off by a jump
crossing
cut point (polyline km 22.3)
second terminal (polyline km 9.7)
cut point (polyline km 22.3)
jump
km into the round trip
0
5
10
15
20
25
6:00
7:00
8:00
9:00
10:00
11:00
12:00
13:00
14:00
15:00
16:00
17:00
18:00
19:00
20:00
21:00
22:00
time of day